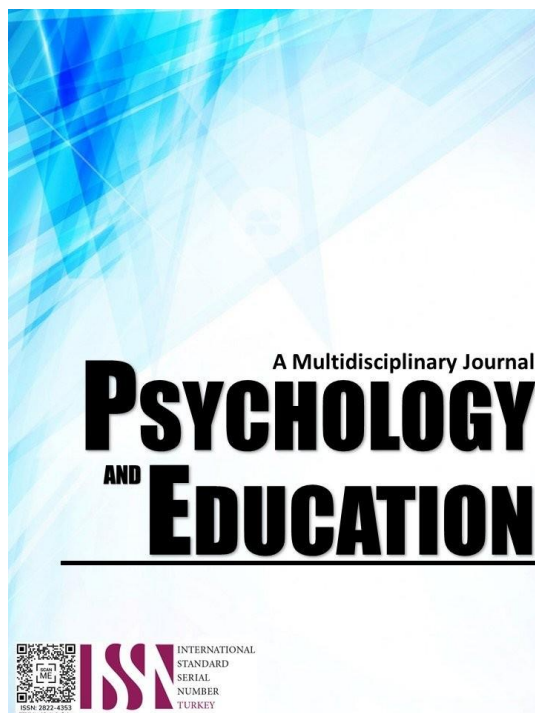


DYNAMIC LEARNING PROGRAM AND ITS EFFECT ON STUDENTS' MATHEMATICS ACHIEVEMENT



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Dynamic Learning Program and its Effect on Students' Mathematics Achievement

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Abstract

This study sought to determine the effects of Dynamic Learning Program (DLP) to the mathematics achievement of Grade 11 students in Alugan National School of Craftsmanship and Home Industries. Quantitative research method and quasi-experimental design were utilized in this study. Two sections of Grade 11 students participated in this study as part of the control group and experimental group, each composed of twenty-three (23) students. The control group was taught using traditional method and the experimental group was taught using DLP. The performance of two groups were measured and compared using a researcher-made pretest and posttest. Results revealed that the use of dynamic learning program in teaching mathematics can improve students' learning. Although learning took place in both groups taught using the traditional method and DLP, the students who are exposed to dynamic learning program have higher learning gains and performed better than the students who are taught using traditional method.

Keywords: *effectiveness; dynamic learning program; experimental*

Introduction

Mathematics has been considered as a fundamental academic subject in school since it develops the arithmetic, logical, and critical reasoning of an individual. It involves several aspects of human life from complex to simple things such as budgeting (Rossi et al., 2023). It has a role in enhancing students' knowledge and skills so they can handle daily problems, pursue higher education, and become productive workers (Maamin et al., 2021). For this account, educational institutions seek to enhance the students, problem-solving and computational skills. Needless to say, mathematics has long been recognized as the backbone of any country's scientific, technological, and economic growth.

Problem in mathematics performance of the students gradually became a serious matter globally. The Organization for Economic Cooperation and Development (OECD) released the results of its 2022 Programme for International Student Assessment (PISA) showing that during the period from 2018 to 2022, the average performance of OECD countries decreased by a record 15 points (Harper, 2023). This showed an alarming threat of continuous downfall of mathematics performance globally in the future. In the United States, the mathematics performance of the teenagers has declined significantly since 2018, which is considered to be lower than their performance 20 years ago (Mervosh, 2023). Meanwhile, Europe experienced its unprecedented drop in mathematics achievement. PISA 2022 results implied that European countries have lagged behind in terms of academic standards (Harper, 2023). Meanwhile, although six of the Asian countries namely Singapore, Chinese Taipei, Macao, Hong Kong, Japan, and Korea performed well, it wasn't able to outshine the poor performance of other countries such as the Philippines.

In the Philippines, the mathematics performance of the students has become a massive challenge in the educational sector of the government. PISA 2022 results stated that in mathematical literacy, Filipino students scored 355 points on average, much lower than the OECD average. It was revealed that just about 1 out of 5 Filipino students (16%) attained at least the minimum proficiency level (Level 2) in mathematical literacy. Meanwhile, the Philippines scored 297 and 249 in mathematics and science, respectively, in the 2019 Trends in International Mathematics and Science Study (TIMSS), the lowest among the 58 countries studied. This results only showed that Filipino students has lagged behind numerous countries when it comes to mathematics learning. Researches have shown that Filipino students have unsatisfactory or poor performance in mathematics (Aguhayon et al., 2023; Azucena et al., 2022; Pentang et al., 2021).

Several studies have been conducted to determine the main reason of such problem and to alleviate this crisis. Upon extensive review of existing literature, the factors that influence the mathematics performance of the students are categorized into three: student-related, teacher-related, and school-related. The student-related factors include mathematics skills, attitude, math anxiety, study habits, self-efficacy, and study orientation of the students (Callaman & Itaas, 2023; Guinocor et al., 2020; Landicho, 2021; Tamayo, 2021; Tak et al., 2022; Rossi et al., 2023). Brezavšek et al. (2020) mentioned in their study that some of the factors that influence students' mathematics performance are the students' attitude and anxiety towards mathematics, their engagement in learning activities, and their attitude towards the utilization of technology in the learning materials. Aside from student-related factors, researchers also identified a number of teacher-related factors that affect mathematics performance. One of these major factors include the teaching strategies of the mathematics teachers (Copioso et al., 2023; Ganál & Guiab, 2023). These studies indicated that teachers should continuously equip themselves with varied teaching strategies in teaching different topics in mathematics. Meanwhile, lesson planning plays an important role in preparing effective teaching strategies in teaching mathematics. In fact, a study of Landas and Alora (2022) encouraged the practice of localized lesson planning to promote higher rate of comprehension in mathematics lessons. Additionally, school-related factors such as the environment and programs implementation has been identified to be factors affecting mathematics performance. Ganál and Guiab (2023) mentioned in their study that a significant role of teachers is to shape the school environment in such a way that it will motivate students to learn mathematics. As part of the school environment, social factors such as peer support, teacher

support, and parent support also contribute to the mathematics learning of the students. Specifically, teacher support and engagement influence the students' motivation in learning mathematics (Doño & Maugila, 2021). Another school-related factor is the program implementation. In fact, a study conducted by Chand et al. (2021) discovered that poor performance in mathematics was caused by an inefficient mathematics program in secondary schools.

The problem posed by the alarming results in international surveys in mathematics performance of students, organizations led the mathematics educators in searching for the root of the problem. Aside from seeking for the reasons about the problem in mathematics performance, educators also suggested teaching strategies and programs that would increase the performance of the learners. Some of these strategies are in the form of learning approaches such as differentiated instruction, gamification, and problem-solving approach (Magayon & Tiu, 2020; Antonio & Tamban, 2022; Albay, 2020). Math educators also tried to apply innovations just to minimize the problem in mathematics performance. Yabo (2020) has provided an innovative learning approach named joyful learning to boost learner participation in the learning process. Meanwhile, Garzon and Casinillo (2021) suggested the use of block models for strategic problem-solving in mathematics. The use of manipulatives in teaching mathematics has also been recognized as an effective approach as shown in the study of Abarquez (2020). This goes to show the efforts of math educators in battling the existing problem in mathematics performance of students.

Aside from the aforementioned learning strategies, Dynamic Learning Program (DLP) introduced by Dr. Christopher Bernido and Dr. Ma. Victoria Bernido has been perceived also to be a way in combating poor mathematics performance. DLP is a learner-centered approach that improves students' critical and creative thinking through innovative learning strategies (Evardo, 2023). According to the study of Evardo (2023), students in the DLP use comprehensive portfolios instead of standard notebooks. These color-coded portfolios for several academic areas serve as a summary of all activities, quizzes, and tests done in school. Students are not permitted to take their portfolios home, fostering independence and critical thinking. The absence of notes encourages learners to participate. Deep thinking promotes learning through making sense of concepts, issues, questions, and ideas encountered during school activities. The full portfolio becomes a significant source of information, tracking a student's development and allowing for self-evaluation, reflection, and efficient filing. Portfolios, which are accessible to both teachers and students, promote cooperative assessment and form a collaboration in the learning process. The study also showed that there is a significant relationship between student perception in DLP and their performance in mathematics.

The DLP is a learner-centered approach that promotes learning-by-doing and active participation of the students. This is consistent with the goals of Constructivism which states that knowledge is not just received from an external reality but rather constructed by the individual (Lowenthal & Muth, 2008). This approach is also supported by the ideals of a Pragmatist, John Dewey, who advocates experiential learning. Bruner (1973) as cited by Amineh and Asl (2015), states that students develop new ideas and information based on their existing knowledge as part of the social process of learning.

Studies about Dynamic Learning Program (DLP) have already been conducted focusing mainly on the perceptions of students and teachers towards it. Thus, this study aims to find out its effectiveness in improving the mathematics achievement of the students in Alugan National School of Craftsmanship and Home Industries.

Research Questions

This study was conducted to determine the effect of utilizing dynamic learning program to the mathematics achievement of Grade 11 students. Specifically, it sought to answer the following questions:

1. What is the pretest score of Grade 11 learners in both control and experimental group before teaching the topics using traditional method and DLP, respectively?
2. What is the pretest score of Grade 11 learners in both control and experimental group after teaching the topics using traditional method and DLP, respectively?
3. What is the learning gain score from pretest and posttest of both groups of Grade 11 learners?
4. Is there a significant difference between the pretest and posttest mean scores of both groups of Grade 11 students?
5. Is there a significant difference between the learning gain scores of both groups of Grade 11 students?

Methodology

Research Design

This study utilized quasi-experimental method of research because the groups were not randomly selected. This design was composed of pre-test, experimentation, and post-test. The researcher purposively assigned the control and the experimental groups. In this study, the control group was taught using the traditional method and the experimental group was taught using the Dynamic Learning Program. Both groups were exposed to the same topics in General Mathematics. The researcher selected Alugan National School of Craftsmanship and Home Industries as the locale of the study.

Participants

The researcher used purposive sampling in identifying the participants of this study. Two heterogeneous sections in Grade 11 were

chosen as the control and experimental group of the study. Each group was composed of 23 respondents.

Instruments

Researcher-made instrument was used as pretest and posttest questionnaires. The pretest and posttest questionnaires are composed of 30-items multiple choices tests from the second quarter topics in General Mathematics based on DepEd K to 12 Curriculum Guide. To establish the validity of the instruments, the researcher consulted two teachers in mathematics and a master's degree holder. Each of them assessed the research instruments' face and content validity. A list of objectives for the lessons was prepared and a table of specification (TOS) was constructed. The constructed researcher-made instruments were administered to the Grade 12 students for Item Analysis since they have already taken the subject in the previous year. After series of revisions, the researcher tested the reliability of the research instruments through a pilot testing. The non-participating Grade 11 students were selected to be the pilot-testing group. Pretest and posttest questionnaires were administered twice at different times. A Cronbach's alpha of 0.83 showed high reliability of the questionnaires.

Procedure

Pretest and posttest were administered to the control group and experimental group. The second quarter topics in general mathematics were taught to the control group using traditional method, and the same topics were also given to the experimental group using Dynamic Learning Program (DLP). In traditional method, the teacher used conventional teaching strategies in teaching the students. Meanwhile in DLP, the teacher acted as a learning facilitator, guiding on the side rather than being a sage on the stage. In this learning program, students were provided with series of concept notes and activities in each topic. Forty (40) minutes were given to the students to accomplish the various learning activities independently while twenty (20) minutes were given for them to clarify vague information to the teacher. At the end of the second quarter, the posttest was administered to both groups.

Data Analysis

After gathering the data, the researcher utilized descriptive statistics such as mean and standard deviation in analyzing the pretest and posttest scores of the control group and experimental group. Meanwhile, t-test for dependent samples was used to test if there is a significant difference between the respective pretest and posttest mean scores of the control and experimental groups. In addition, t-test for independent samples was used to test the significant difference between the learning gains of both groups. Moreover, Statistical Package for Social Sciences (SPSS) software was utilized in data analysis.

Ethical Considerations

Consent from the school principal, teachers, parents, and students were secured before the conduct of the study. All of the concerned individuals were properly oriented on the objectives and the procedures. Confidentiality of the information provided by the participants was also emphasized ensuring all data gathered are to be used solely for this research study. Furthermore, no conflict of interest aroused in the conduct of this study.

Results and Discussion

This section provides the information obtained from the control and experimental groups using pretest and posttest. The presentation of the findings follows the problems stated by the study.

Table 1. *Pretest scores of the control and experimental groups*

Score	Description	Control Group		Experimental Group	
		F	%	F	%
25 to 30	Excellent	0	0	0	0
19 to 24	Very Satisfactory	0	0	2	8.7
13 to 18	Satisfactory	7	30.43	9	39.13
7 to 12	Fair	15	65.22	10	43.47
1 to 6	Poor	1	4.35	2	8.7
Total		23	100	23	100
Mean		12.96		13.22	
Standard Deviation		3.637		3.931	

Table 1 shows the pretest performance of the control group and experimental group. Most of the students in both groups have fair or satisfactory performance. This means that most of their scores range from 7 to 18 points for the pretest. However, 2 students from the experimental group have very satisfactory performance. Furthermore, data revealed that the two groups have nearly equal knowledge about the topics in general mathematics which is evident in the mean scores 12.96 and 13.22 for control and experimental group respectively. In addition, the scores in the control group is less scattered than the experimental group as shown by their standard deviations, 3.637 and 3.931 respectively. These figures revealed that there is not much difference between the pretest performance of the control group and experimental group.

Table 2. *Posttest scores of the control and experimental groups*

Score	Description	Control Group		Experimental Group	
		F	%	F	%
25 to 30	Excellent	0	0.00	2	8.70
19 to 24	Very Satisfactory	11	47.83	14	60.87
13 to 18	Satisfactory	8	34.78	5	21.74
7 to 12	Fair	4	17.39	2	8.70
1 to 6	Poor	0	0.00	0	0.00
Total		23	100	23	100
Mean		18.70		19.30	
Standard Deviation		5.085		4.436	

Meanwhile, table 2 shows the posttest performance of the control group and experimental group. Most of the students in both groups have very satisfactory performance in the posttest. This means that most of their scores range from 19 to 24 points for the posttest. In addition, 2 students from the experimental group have excellent performance scoring 25 to 30 points. These data revealed that the students in two groups have nearly equal performance in general mathematics which is evident in the mean scores 18.70 and 19.30 for control and experimental group respectively. However, it was important to note that the scores in the experimental group is less scattered than the control group as shown by their standard deviations, 4.436 and 5.085 respectively.

Table 3. *Learning gain scores in percentage from pretest and posttest of control and experimental groups*

Learning Gain in Percentage	Control Group		Experimental Group	
	F	%	F	%
51-60	0	0	8	34.78
41-50	1	4.35	1	4.35
31-40	5	21.74	2	8.70
21-30	10	43.48	6	26.09
11 to 20	2	8.70	4	17.39
1 to 10	4	17.39	2	8.70
0	1	4.35	0	0.00
Total	23	100	23	100
Mean	25.00		35.48	
Standard Deviation	15.75		12.97	

The learning gains in percentage of the two groups revealed that the experimental group got the highest mean score of 35.48% while the mean score of the control group was 25%. The standard deviations 15.75 and 12.97 for control and experimental group, respectively, showed that the learning gains of the experimental group are less scattered. In the experimental group, there were 8 students or 34.78% who got 51-60 percentage in learning gain, which was the highest for both groups. Meanwhile, the most scored learning gain in the control group was 21-30 percentage in which 10 of the students or 43.48% of the control group achieved. Furthermore, all of the students in the experimental group have significant learning gains, while there is a student in control group who do not have a learning gain. This further emphasized that the DLP applied in the experimental group has been more effective than the traditional method applied in the control group.

Table 4. *Test of difference between the pretest and posttest mean scores of control and experimental groups*

	Mean Scores		Mean Difference	t	Sig. (2-tailed)	Findings
	Pretest	Posttest				
Control	12.96	18.70	5.739	-4.403	.000	Significant
Experimental	13.22	19.30	6.087	-4.925	.000	Significant

Meanwhile, table 4 revealed the test of difference between the pretest and posttest mean scores of the students who are taught using traditional method and those who are exposed to DLP. It showed the mean differences of 5.739 and 6.087 for control and experimental groups respectively. The result revealed that there is a significant difference between the pretest and posttest mean scores of the control group since the computed t-value is -4.403 and a p-value of .000 which is lesser than 0.05 level of significance. Thus, the null hypothesis was rejected.

On the other hand, there is also a significant difference between the pretest and posttest mean scores of the students who are exposed to DLP. The result revealed a t-value of -4.925 with p-value of .000 which is lesser than the 0.05 level of significance. Thus, the null hypothesis was also rejected.

Moreover, table 5 revealed the test of difference between learning gain scores in percentage of the students who are taught using traditional method and those who are exposed to DLP. It showed the mean learning gains of 25% and 35.46% for control and experimental groups respectively. The result revealed that there is a significant difference between the learning gain scores in

percentage of the control group and the experimental group since the computed t-value is -2.463 and a p-value of 0.018 which is lesser than 0.05 level of significance. Thus, the null hypothesis was rejected.

Table 5. *Test of difference between the learning gain scores in percentage of control and experimental groups*

	Mean Gain in %	Standard Deviation	t	Sig. (2-tailed)	Findings
Control	25.00	15.75	-2.463	0.018	Significant
Experimental	35.48	12.97			

The researcher utilized two teaching strategies in this study- traditional method and DLP. Traditional method of teaching uses conventional approaches in teaching mathematics while DLP promotes student-centered learning process. Lankford (2021) mentioned in their study that in student-centered classrooms, teachers and students work together to acquire new knowledge. In DLP, students were also given chance for self-assessment since their activities are compiled in a portfolio providing them an opportunity to monitor their progress. Barana et al. (2022) stated that self-assessment is a technique that encourages students to actively participate in evaluating their own performance.

Results of this study implied that the students in two groups have nearly equal knowledge about the topics in general mathematics. This further denote that students rely on their past knowledge in interpreting newly encountered information. Low scores in the pretest can be considered normal, after all, the respondents of the study have no or little knowledge about the new lessons. This means that the students have used their background knowledge and learned experiences in answering the pretest which is associated on the idea of Schema Theory. This theory states that prior experiences or background information comprise a schema. The theory highlights how people's existing knowledge and knowledge structure have a significant impact on their cognitive processes (Alawdi, 2023). Similarly, Surur (2020) mentioned in their study about Thorndike's Law of Response by Analogy which implies that a person's natural inclinations to react as well as aspects from similar situations in which he has previously developed reactions influence how he will react in a new situation.

In addition, the results also imply that learning takes place even in the traditional method of teaching mathematics similar on how the use of DLP allowed the learners to learn the topics discussed. However, the result emphasized that the students who are exposed to DLP have higher learning gains than those students who are taught using traditional method. Nevertheless, it was undeniable that both traditional method and DLP are effective in teaching mathematics. This case is similar to the study of Baker (2018) as cited by Hafeez (2021) when they compare the effectiveness of traditional teaching method and a student-centered learning approach which is blended learning. It was found out that both teaching strategy develop the same learning achievement. With this, Alessa and Hussein (2023) cited that Mahta (2019) stressed out that teachers should be able to know when a traditional method will work and when to innovate learning approaches.

Furthermore, it is important to note that the 35.48% mean learning gain of the experimental group compared to the 25% of the control group implied that the learning gains of the students who are exposed in DLP is higher than that of the students who are taught using traditional method. The results further denote that the students exposed in DLP learn better than the students who are taught with traditional method. This gives a notion that student-centered approach is more effective than teacher-centered approach. Emanet and Kezer (2021) concluded in their study that when it comes to math achievement, the student-centered approaches outperform the teacher-centered ones. This shows the need for teachers to innovate their teaching strategies in order to promote meaningful learning in mathematics and motivate students to actively participate in the learning process. Sugano et al. (2019) as cited by Mamolo and Sugano (2020) emphasized that in order to make the necessary pedagogical and instructional improvements, teachers must work harder to deliver high-quality instruction and implement new procedures for evaluating their teaching. Moreover, Diamante and Banca (2021) stressed out that in order to adapt their instruction to the different learning styles of their students, teachers should use a variety of teaching strategies. Instead of the typical teacher-centered approach to instruction, teachers should move toward a more student-centered approach where students take responsibility of their learning (Bature, 2020).

Conclusions

From the aforementioned findings of the study, it can be concluded that the use of Dynamic Learning Program (DLP) in teaching mathematics can improve students learning. The idea that the teacher acts as a facilitator rather than a center of attention proves to be essential in promoting meaningful mathematics learning to the students. Although learning took place in both groups taught using the traditional method of teaching and DLP, the students who are exposed to DLP performed better than the students who are taught using traditional method. Thus, it can be concluded that DLP improves students' mathematics achievement. Moreover, mathematics teachers should utilize dynamic learning program or other teaching strategies where students do the work while the teacher acts as facilitator.

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